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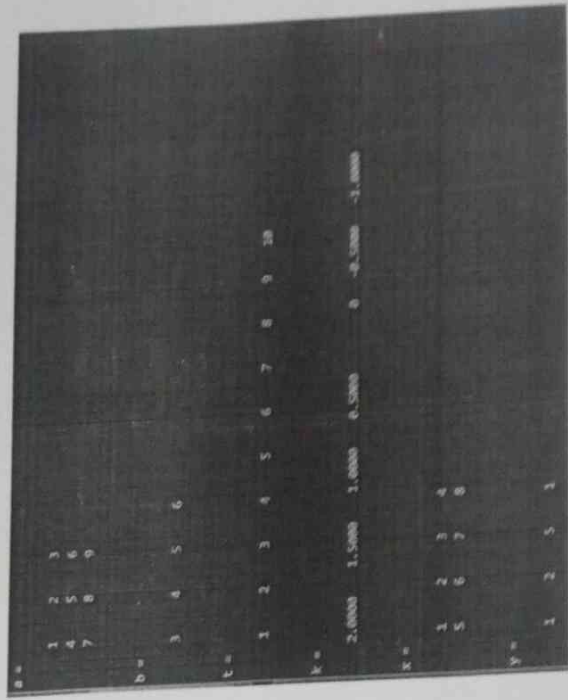
Aim: Introduction to the MATLAB commands

Practical No. 1

- `clc` : Clears command window
- `clear` : Removes the variables from memory
- `cd` : Changes the current directory
- `date` : Displays current date
- `delete` : Deletes a file
- `find` : Finds the indices of nonzero elements
- `max` : Returns the largest element
- `min` : Returns the smallest element
- `plot` : Generates xy plot
- `title` : Puts title at the top of the plot
- `linspace` : creates regularly spaced vector
- `zeros` : Creates a matrix or array filled with zeros of size m-by-n.
- `Ones` : Creates a matrix or array filled with ones of size m-by-n.
- `reshape` : Reshapes a matrix or array into a new size specified by m and n.
- `subplot` : Creates plots in subwindows.
- `hold` : Hold the current graph, see also figure

Practical No. 2
Aim: Write a program to represent arrays in different ways

- `a=[1 2 3; 4 5 6; 7 8 9]`
- `b=[3 :6]`
- `t=1:10`
- `k=2:-0.5:-1`
- `x=[1:4; 5:8]`
- `y=[1 2 5 1]`



Practical No. 3

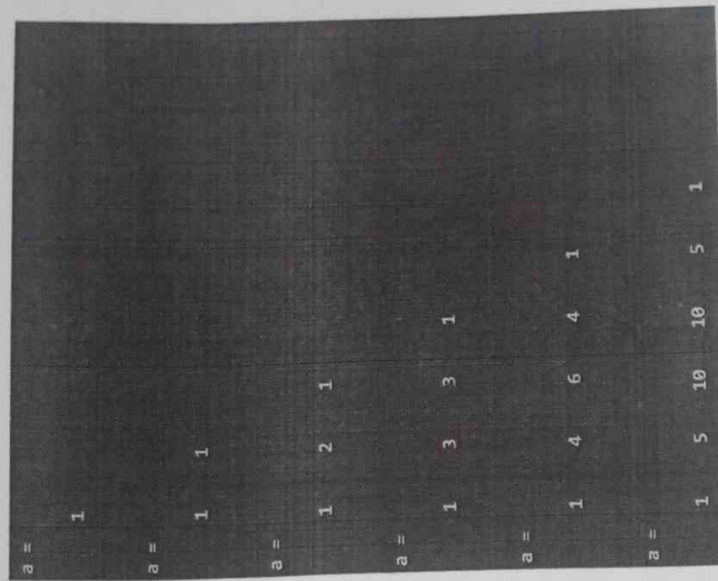
Aim: Write a program to perform various matrices arithmetic operation

- $a = [1 \ 2 \ 3; 4 \ 5 \ 6; 7 \ 8 \ 9]$
- $b = [3 \ 4 \ 5; 1 \ 4 \ 5; 3 \ 5 \ 6]$
- $\text{sum} = a + b;$
- $\text{sub} = a - b;$
- $\text{mult} = a * b;$

```
a =  
     1     2     3  
     4     5     6  
     7     8     9  
  
b =  
     3     4     5  
     1     4     5  
     3     5     6  
  
>>
```

Practical No. 4
Aim: Write a program to print out Pascal's triangle:

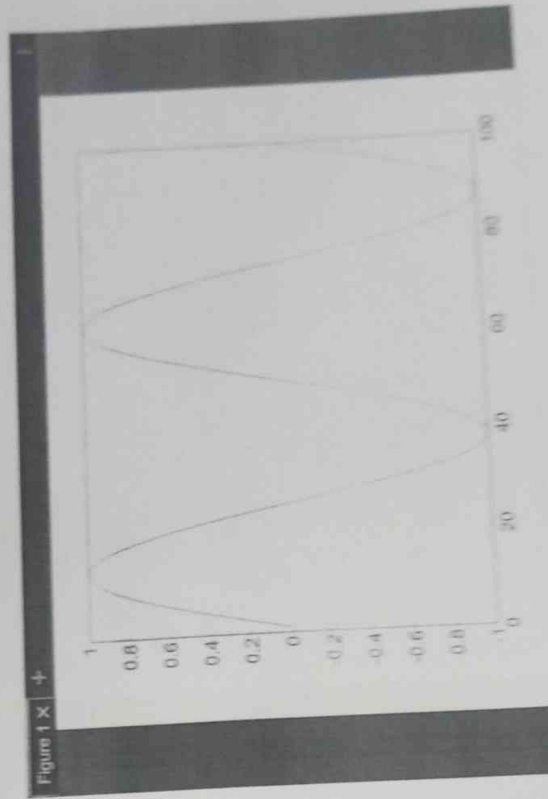
- a = 1
- while length(a) < 10
- a = [0 a] + [a 0]
- end



Practical No. 5

Aim: Write a program Plot the function $\sin(x)$ between $0 \leq x \leq 4\pi$ using linspace

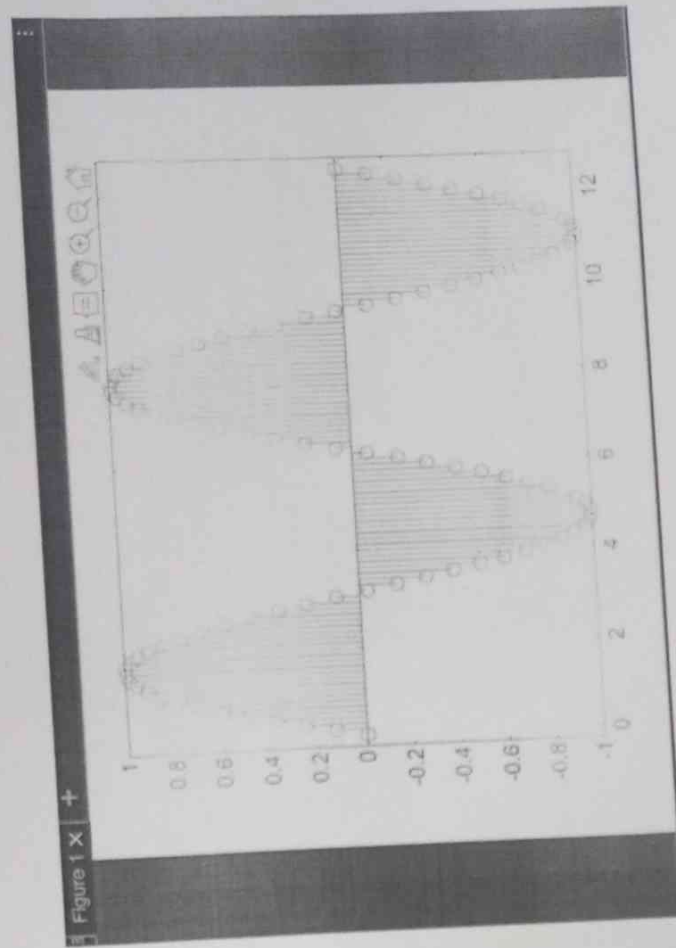
- `x=linspace(0,4*pi,100);`
- `y=sin(x);`
- `plot(y);`



Practical No. 6

Aim: Write a program Plot the function $\sin(x)$ between $0 \leq x \leq 4\pi$ using stem function

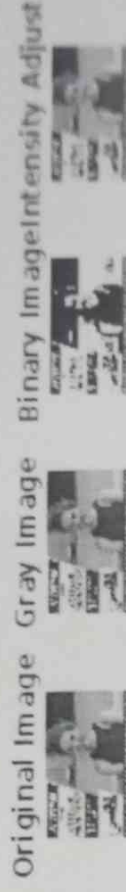
- `x=linspace(0,4*pi,100);`
- `y=sin(x);`
- `stem(y);`
- `stem(x,y);`



Practical No. 7

Aim: Write a program to import Digital Images and convert the class of image into gray, and binary, also adjust the intensity levels

```
a = imread('peppers.png');
subplot(1,4,1);
imshow(a);
title('Original Image');
b = rgb2gray(a);
subplot(1,4,2);
imshow(b);
title('Gray Image');
c = im2bw(a);
subplot(1,4,3);
imshow(c);
title('Binary Image');
d = imadjust(b);
subplot(1,4,4);
imshow(d);
title('Intensity Adjust');
```

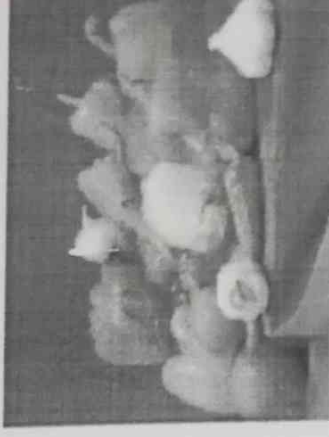


Practical No. 8

Aim: Write a program to import Digital Images and reduce the size of the image
(50%)

- `I = imread('peppers.png');`
- `J = imresize(I, 0.5);`
- `Subplot(1,2,1);`
- `imshow(I);`
- `title('Original Image');`
- `Subplot(1,2,2);`
- `imshow(J);`
- `title('Resize Image');`
- `sz1=size(I);`
- `sz2=size(J);`

Original Image



Resize Image



Practical No. 9

AIM: Write a program to import Digital Images and perform arithmetic operations

```
clear
close all;
a=imread('cameraman.tif');
b=imread('rice.png');
c=a+b; % addition
subplot(1,3,1);
imshow(c);
title('ADD. of Image 1 & 2');
d=a-b; % subtraction
subplot(1,3,2);
imshow(d);
title('SUB. of Image 1 & 2');
e=a.*b; % multiplication
subplot(1,3,3);
imshow(e);
title('Mult. of Image 1 & 2');
```

ADD. of Image 1 & 2 SUB. of Image 1 & 2



Practical No. 10

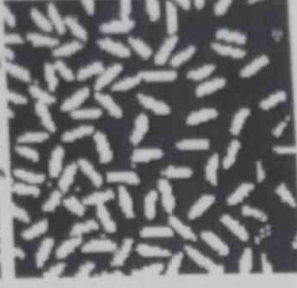
AIM: Write a program to import Digital Images and perform logical AND operations

- `a= imread('cameraman.tif');`
- `b= imread('rice.png');`
- `a1=im2bw(a);`
- `Subplot(1,3,1);`
- `imshow(a1);`
- `title('Binary Image a');`
- `b1=im2bw(b);`
- `Subplot(1,3,2);`
- `imshow(b1);`
- `title('Binary Image b');`
- `C=and(a1,b1);`
- `Subplot(1,3,3);`
- `imshow(c);`

Binary Image a



Binary Image b



AND Image

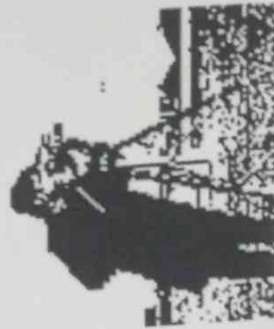


Practical No. 11

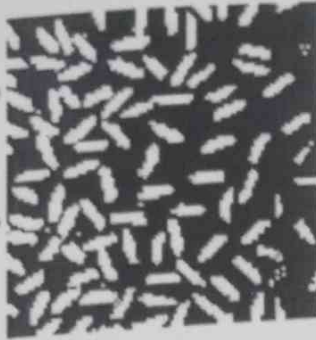
AIM: To import Digital Images and perform logical OR operations

- `a= imread('cameraman.tif');`
- `b= imread('rice.png');`
- `a1=im2bw(a);`
- `Subplot(1,3,1);`
- `imshow(a1);`
- `title('Binary Image a');`
- `b1=im2bw(b);`
- `Subplot(1,4,2);`
- `imshow(b1);`
- `title('Binary Image b');`
- `C=or(a1,b1);`
- `Subplot(1,3,2);`
- `imshow(c);`
- `title('OR Image');`

Binary Image a



Binary Image b



OR Image



Practical No. 12

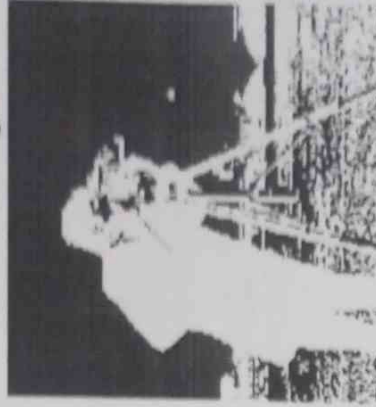
AIM: To import Digital Images and perform logical NOT operations

- `a= imread('cameraman.tif');`
- `a1=im2bw(a);`
- `subplot(1,2,1);`
- `imshow(a1);`
- `title('Binary Image a');`
- `C=not(a1);`
- `subplot(1,2,2);`
- `imshow(C);`
- `title('NOT Image');`

Binary Image a



NOT Image



Practical No. 13

AIM: To Select a Digital Image and perform Basic Intensity Transformation "Image Negative"

```
I1=imread('tree.jpg');  
Subplot(1,2,1);  
Imshow(i1);  
I2=i1;  
For r =1:256  
    For c =1:256  
        I2(r,c)=255-I2(r,c);  
    End  
End  
Subplot(1,2,2);  
Imshow(i2);
```



Practical No. 14

AIM: To Select a Digital Image and perform Basic Intensity Transformation "Logarithmic Transformation"

```
img = imread('Sampleimage.jpg');
```

```
r = double(img);
```

```
C = 1;
```

```
S = C * log(1 + r);
```

```
T = 255 / (C * log(256));
```

```
B = uint8(T * S);
```

```
imshow(B);
```

```
title('Log Transformation');
```



Practical No. 15

AIM: To Select a Digital Image and perform Basic Intensity Transformation "Power Law Transformation"

```
img = imread('Sampleimage.jpg');  
subplot(1,2,1);  
imshow(img);  
r = double(img);  
G = 0.6;  
S = C * (r.^G);  
T = 255/(C * (255.^G));  
O = uint8(T * S);  
subplot(1,2,2);  
imshow(O);  
title('Power Law Transformation');
```

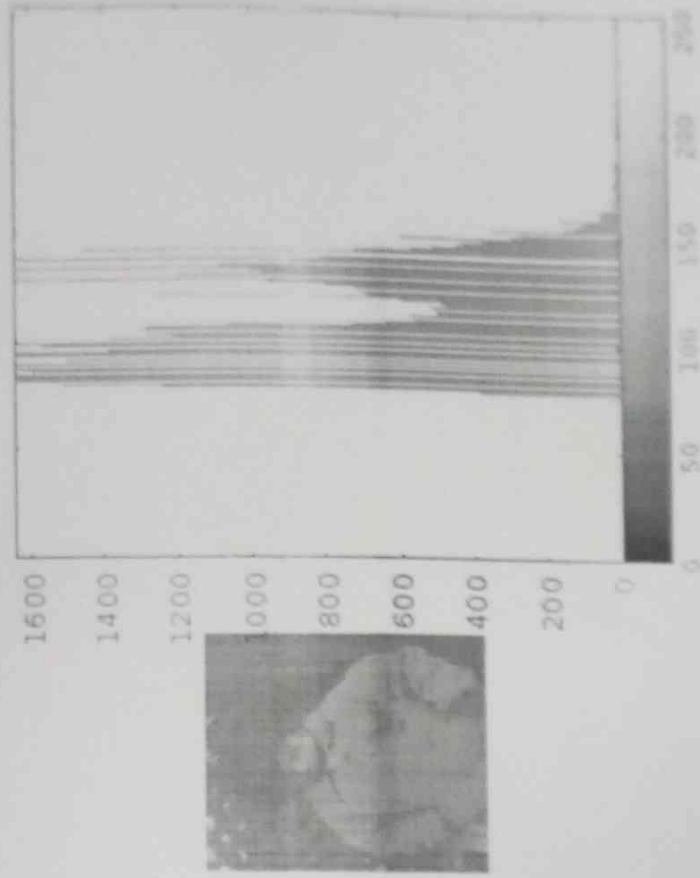
Power Law Transformation



Practical No. 16

Aim: To Select a Digital Image and Plot a Histogram

- `I = imread("pout.tif");`
- `subplot(1,3,1);`
- `imshow(I);`
- `subplot(1,3,2:3);`
- `imhist(I)`



Practical No. 17

Aim: To Select a Digital Image, perform Histogram Equalization and plot the histogram.

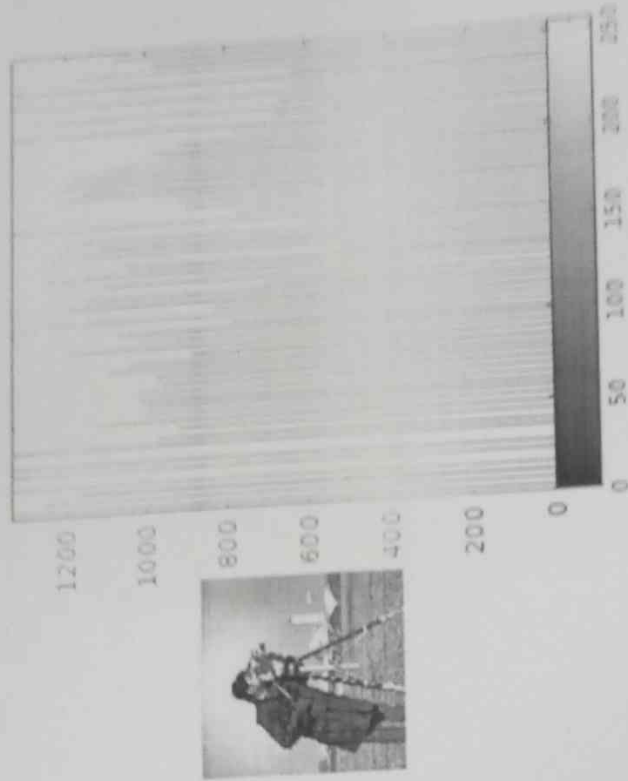
- `I = imread("abc.tif");`
- `subplot(1,3,1)`
- `imshow(I)`
- `subplot(1,3,2:3)`
- `imhist(I)`
- `J = histeq(I);`
- `subplot(1,3,1)`
- `imshow(J)`
- `subplot(1,3,2:3)`
- `imhist(J)`



Practical No. 17

Aim: To Select a Digital Image, perform Histogram Equalization and plot the histogram.

- `I = imread("abc.tif");`
- `subplot(1,3,1)`
- `imshow(I)`
- `subplot(1,3,2:3)`
- `imhist(I)`
- `J = histeq(I);`
- `subplot(1,3,1)`
- `imshow(J)`
- `subplot(1,3,2:3)`
- `imhist(J)`



Practical No. 18

Aim: To import a Digital Image and apply an image averaging filter.

- `I = imread('cameraman.tif');`
- `subplot(1, 2, 1);`
- `imshow(I);`
- `title('Original Image');`
- `C=fspecial('average',[3,3]);`
- `d=imfilter(I,C);`
- `subplot(1, 2, 2);`
- `imshow(d);`
- `title('3x3 average filter');`

Original Image



3x3 Average Filter



Practical No. 19

Aim: To import a Digital Image and apply a median filter.

- `I = imread('cameraman.tif');`
- `subplot(1, 3, 1);`
- `imshow(I);`
- `title('Original Image');`
- `J = imnoise(I, 'salt & pepper', 0.02);`
- `subplot(1, 3, 2);`
- `imshow(J);`
- `title('Noise Image');`
- `K = medfilt2(J);`
- `subplot(1, 3, 3);`
- `imshow(K);`
- `title('Filtered image');`

Original Image



Noisy Image



Filtered Image



Practical No. 20

Aim: To import a Digital Image and apply a min and max filter.

- `Original=imread('cameraman.tif');`
- `BW = im2bw(Original,0.6);`
- `minf=@(x) min(x(:));`
- `maxf=@(x) max(x(:));`
- `min_Image=nfilter(BW,[3 3],minf);`
- `max_Image=nfilter(BW,[3 3],maxf);`
- `subplot(2,2,1);`
- `imshow(BW);`
- `title('Original');`
- `subplot(2,2,2);`
- `imshow(min_Image);`
- `title('Min');`
- `subplot(2,2,3);`
- `imshow(max_Image);`
- `title('Max');`

Original



Min Filter



Max Filter



Practical No. 21

Aim: To import a Digital Image and apply a Laplacian filter.

- `A = imread('peppers.png');`
- `sigma = 0.4;`
- `alpha = 0.5;`
- `B = locallapfilt(A, sigma, alpha);`
- `imshowpair(A, B, 'montage')`

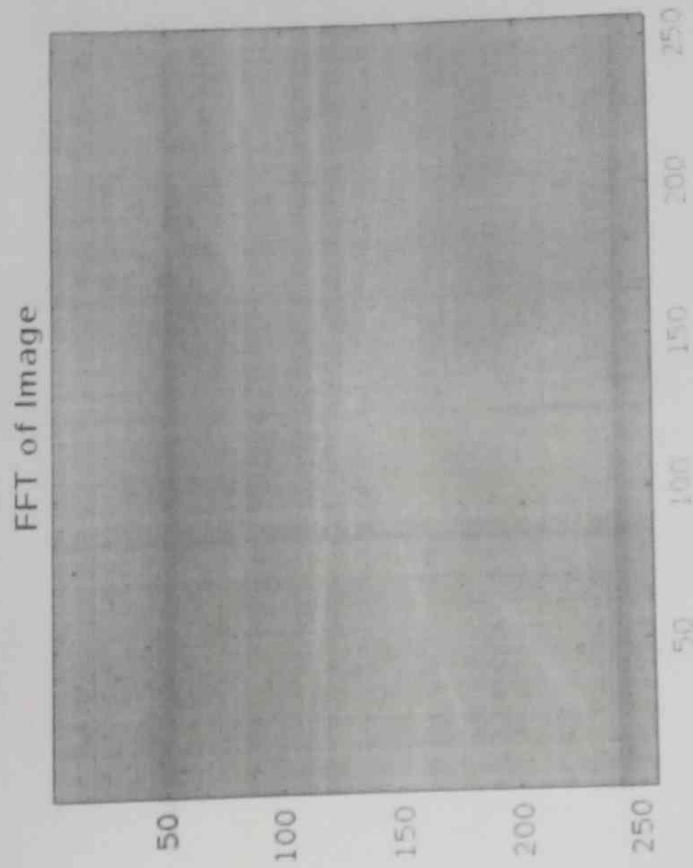
Original and Laplacian Filtered Images



Practical No. 22

Aim: To import a Digital Image and apply a FFT.

- `I=imread('cameraman.tif');`
- `F=fft2(double(I));`
- `S=fftshift(F);`
- `L=log2(S);`
- `A=abs(L);`
- `imagesc(A)`



Practical No. 23

Aim: To import a Digital Image and apply a frequency domain filter.

- `I = imread("cameraman.tif");`
- `I = im2double(I);`
- `F = fspecial("average",3);`
- `Ipad = padarray(I,[3-1 3-1],0,"post");`
- `Fpad = padarray(F,[256-1 256-1],0,"post");`
- `Ifft = fft2(Ipad);`
- `Ffft = fft2(Fpad);`
- `Offt = Ifft.*Ffft;`
- `Opad = ifft2(Offt);`
- `O = Opad(2:end-1,2:end-1);`

Frequency Domain Filter



Aim: To import a Digital Image and Perform various Morphological operation erosion

- I=imread('cameraman.tif');
- I=im2bw(I);
- se=ones(5, 5);
- [P, Q]=size(se);
- In=zeros(size(I, 1), size(I, 2));
- for i=ceil(P/2):size(I, 1)-floor(P/2)
 - for j=ceil(Q/2):size(I, 2)-floor(Q/2)
 - on=I(i-floor(P/2):i+floor(P/2), j-floor(Q/2):j+floor(Q/2));
 - nh=on(logical(se));
 - In(i, j)=min(nh(:));
- end
- end
- imshow(In);



Practical No. 25
Aim: To import a Digital Image and Perform various Morphological operation dilation



```

• I=imread('cameraman.tif');
• I=im2bw(I);
• se=ones(5, 5);
• [P, Q]=size(se);
• In=zeros(size(I, 1), size(I, 2));
• for i=ceil(P/2):size(I, 1)-floor(P/2)
    • for j=ceil(Q/2):size(I, 2)-floor(Q/2)
        • on=I(i-floor(P/2):i+floor(P/2), j-floor(Q/2):j+floor(Q/2));
        • nh=on(logical(se));
        • In(i, j)=max(nh(:));
    • end
• end
• imshow(In);
    
```

Practical No.26

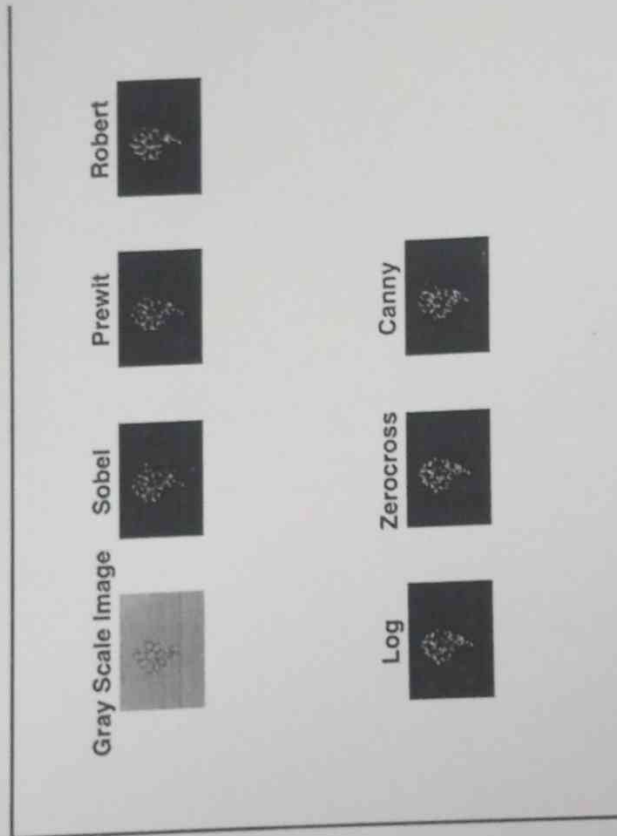
Aim: To perform edge detection of an image.

```
A = imread("Flower.png")
I = rgb2gray(A);
subplot(2,4,1);
imshow(I);
title("Gray Scale Image");
% Sobel Edge Detection
J = edge(I, 'Sobel');
subplot(2, 4, 2),
imshow(J);
title("Sobel");
% Prewitt Edge detection
K = edge(I, 'Prewitt');
subplot(2, 4, 3),
imshow(K);
title("Prewitt");
% Robert Edge Detection
L = edge(I, 'Roberts');
subplot(2, 4, 4),
imshow(L);
title("Robert");
% Log Edge Detection
M = edge(I, 'log');
subplot(2, 4, 5),
imshow(M);
```

```

title("Log");
% Zerocross Edge Detection
M = edge(1, 'Zerocross');
subplot(2, 4, 6);
imshow(M);
title("Zerocross");
% Canny Edge Detection
N = edge(1, 'Canny');
subplot(2, 4, 7);
imshow(N);
title("Canny");

```



Practical No.27

Aim: Edge Detection using Morphological Erosion (Inner Boundary)

```
k = imread('cameraman.tif');  
SE = strel('disk', 2, 0);  
k1 = imerode(k, SE);  
k2 = k - k1;  
imtool(k);  
imtool(k2);
```



Practical No.28

Aim: Edge Detection using Morphological Dilation(Outer Boundary)

```
k = imread("cameraman.tif");  
SE = strel("disk", 2, 0);  
k1 = imdilate(k, SE);  
k2 = k1 - k;  
imtool(k);  
imtool(k2);
```



Practical No.29

Aim: Low-Pass Filtering in Frequency Domain on an Image.

```
A=imread("cameraman.tif")
A1=im2double(A);
y1 = fft2(A1);
[N, M] = size(A1);
u=-1/2:1/M:1/2-1/M;
v=-1/2:1/N:1/2-1/N;
[U, V] = meshgrid(u, v);
d=U.*U/(0.13*0.13) + V.*V/(0.13*0.13);
mask = d <= 1;
yt1 = fftshift(y1);
yt2=yt1.*mask;
LPF = ifft2(ifftshift(yt2), 'symmetric');
figure()
imshow(A)
figure()
imshow(yt1)
figure()
imshow(LPF)
```



Practical No.30

Aim: High pass filter.

```
A=imread('cameraman.tif');  
D = fftshift(fft2(A));  
h1 = (1/9).*[1,1,1;1,1,1;1,1,1];  
B1=filter2(h1,D);  
B1 = uint8(round(B1));  
G=ifft2(B1);  
G2=log(abs(G))  
figure()  
imshow(A)  
figure()  
imshow(G2)
```

